

Differential Calculus

*Home work sheet # 7

Partial derivatives

- Let $f(x, y) = 3x^3y^2$. Find
(a) $f_x(x, y)$, (b) $f_y(x, y)$, (c) $f_x(x, 1)$, (d) $f_y(1, y)$, (e) $f_x(1, 2)$, (f) $f_y(1, 2)$.
- Let $f(x, y) = xe^{-y} + 5y$, $f(x, y) = \sqrt{3x + 2y}$
(a) Find the Slope of the surface $f(x, y)$ in the x- direction at the point (3,0) and (4,2)
(b) Find the slope of the surface $f(x, y)$ in the y direction of the point (3,0) and (4,2)
- Let $f(x, y) = 4x^2 - 2y + 7x^4y^5$, Find (a) f_{xx} , (b) f_{yy} , (c) f_{xy} , (d) f_{yx} .
- Let $z = \sin(y^2 - 4x)$ and $z = (x + y)^{-1}$.
(a) Find the rate of Change of z w. r. to x at the point (2,1) and (-2, 4) with y kept fixed.
(b) Find the rate of Change of z w. r. to y at the point (2,1) and (-2, 4) with x kept fixed.
- Let $f(x, y, z) = x^3y^5z^7 + xy^2 + y^3z$, find
(a) f_{xy} (b) f_{yz} (c) f_{xz} (d) f_{zz} (e) f_{zyy} (f) f_{zxy} (g) f_{zyx} (h) f_{xyz} .
- Let $f(x, y, z) = \sqrt{xy} + \ln(x^2z^3) - x \tan(z)$. Compute f_x , f_z , f_{xy} , f_{xyz} .

Chain Rule

7-10 Use an appropriate form of the chain rule to find dw/dt . ■

7. $w = 5x^2y^3z^4$; $x = t^2$, $y = t^3$, $z = t^5$

8. $w = \ln(3x^2 - 2y + 4z^3)$; $x = t^{1/2}$, $y = t^{2/3}$, $z = t^{-2}$

9. $w = 5 \cos xy - \sin xz$; $x = 1/t$, $y = t$, $z = t^3$

10. $w = \sqrt{1 + x - 2yz^4x}$; $x = \ln t$, $y = t$, $z = 4t$

17-22 Use appropriate forms of the chain rule to find $\partial z/\partial u$ and $\partial z/\partial v$. ■

17. $z = 8x^2y - 2x + 3y$; $x = uv$, $y = u - v$

18. $z = x^2 - y \tan x$; $x = u/v$, $y = u^2v^2$

19. $z = x/y$; $x = 2 \cos u$, $y = 3 \sin v$

20. $z = 3x - 2y$; $x = u + v \ln u$, $y = u^2 - v \ln v$

21. $z = e^{x^2y}$; $x = \sqrt{uv}$, $y = 1/v$

22. $z = \cos x \sin y$; $x = u - v$, $y = u^2 + v^2$

23. Let $T = x^2y - xy^3 + 2$; $x = r \cos \theta$, $y = r \sin \theta$. Find $\partial T/\partial r$ and $\partial T/\partial \theta$.

24. Let $R = e^{2s-t^2}$; $s = 3\phi$, $t = \phi^{1/2}$. Find $dR/d\phi$.

25. Let $t = u/v$; $u = x^2 - y^2$, $v = 4xy^3$. Find $\partial t/\partial x$ and $\partial t/\partial y$.

26. Let $w = rs/(r^2 + s^2)$; $r = uv$, $s = u - 2v$. Find $\partial w/\partial u$ and $\partial w/\partial v$.

27. Let $z = \ln(x^2 + 1)$, where $x = r \cos \theta$. Find $\partial z/\partial r$ and $\partial z/\partial \theta$.

28. Let $u = rs^2 \ln t$, $r = x^2$, $s = 4y + 1$, $t = xy^3$. Find $\partial u/\partial x$ and $\partial u/\partial y$.

29. Let $w = 4x^2 + 4y^2 + z^2$, $x = \rho \sin \phi \cos \theta$,
 $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$. Find $\partial w/\partial \rho$, $\partial w/\partial \phi$, and $\partial w/\partial \theta$.

30. Let $w = 3xy^2z^3$, $y = 3x^2 + 2$, $z = \sqrt{x-1}$. Find dw/dx .

31. Use a chain rule to find the value of $\left. \frac{dw}{ds} \right|_{s=1/4}$ if
 $w = r^2 - r \tan \theta$; $r = \sqrt{s}$, $\theta = \pi s$.

32. Use a chain rule to find the values of
 $\left. \frac{\partial f}{\partial u} \right|_{u=1, v=-2}$ and $\left. \frac{\partial f}{\partial v} \right|_{u=1, v=-2}$
if $f(x, y) = x^2y^2 - x + 2y$; $x = \sqrt{u}$, $y = uv^3$.

33. Use a chain rule to find the values of
 $\left. \frac{\partial z}{\partial r} \right|_{r=2, \theta=\pi/6}$ and $\left. \frac{\partial z}{\partial \theta} \right|_{r=2, \theta=\pi/6}$
if $z = xye^{x/y}$; $x = r \cos \theta$, $y = r \sin \theta$.

34. Use a chain rule to find $\left. \frac{dz}{dt} \right|_{t=3}$ if $z = x^2y$; $x = t^2$, $y = t + 7$.

*These problems are for the students only as home work. Search the reference books for more examples.